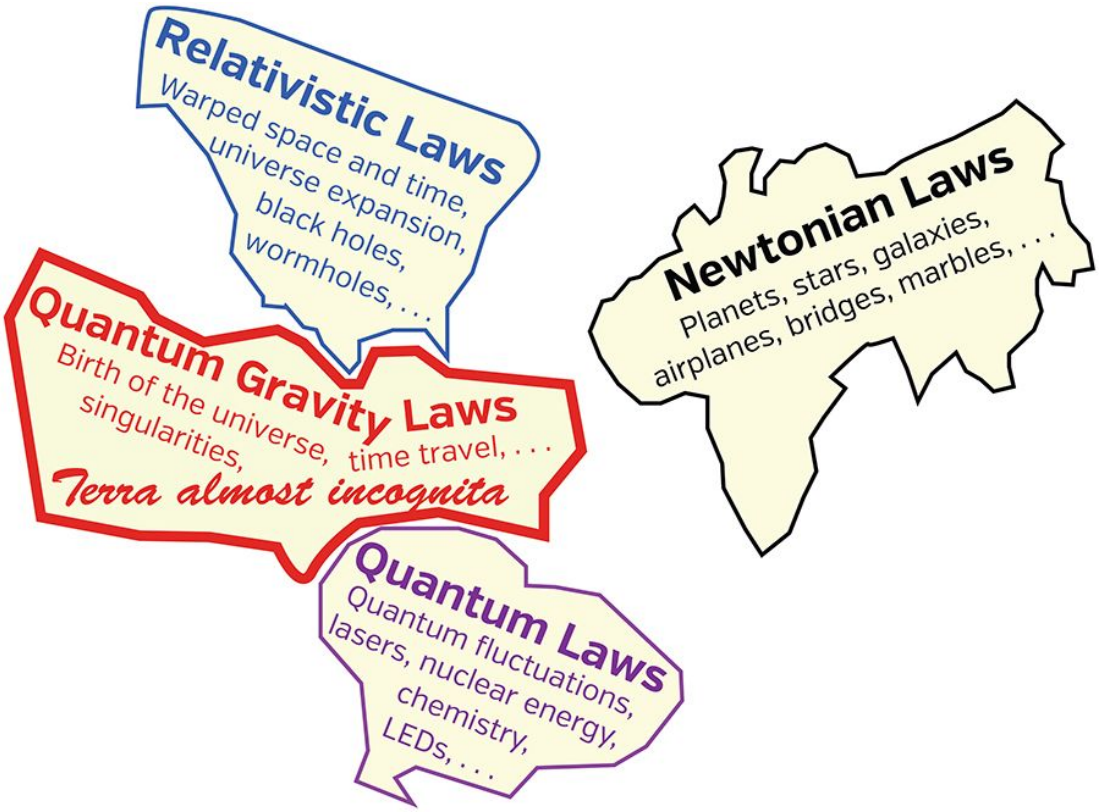


Fig. 3.2. The physical laws that govern the universe.

By 1957 it became evident that the relativistic laws and the quantum laws are fundamentally incompatible. They predict different things, incompatible things, in realms where gravity is intense *and* quantum fluctuations are strong.[5] These realms include the big bang birth of our universe (Chapter 2), the cores of black holes like Gargantua (Chapters 26 and 28), and backward time travel (Chapter 30). In these realms a "fiery marriage"[6] of the incompatible relativistic and quantum laws gives rise to new *laws of quantum gravity* (Figure 3.2).

We do not yet know the laws of quantum gravity, but we have some compelling insights, including superstring theory (Chapter 21), thanks to enormous effort by the world's greatest twenty-first-century physicists. Despite those insights, quantum gravity remains terra almost incognita (an almost unknown land). This leaves much elbow room for